Gaming Mouse SPEC

Date	Version	Content	Author
2021/07/20	1.0	新版 mouse protocol	Easen

軟體規格

1 ． Application program function list:

- ✧8 Profiles in Flash.
- ✧8Macro Stage (G0~G48),each macro 250byte(125 actions) length for all profile.
- ✧Macro Loop= 1~65535.
- ✧All standard keys can change key map (Assign a key).
- ✧Send macro key or send key on boot protocol with USB key code format.
- ✧All standard keys can be configured to fire key
- ✧1/2/4/8mSec report rate.
- ✧Can be set to a maximum of 8 DPI stage

一.Driver Release Package:

- ✧Setup Package
- ✧INF Package (WHQL)

USB Protocol:

1. Use SetFeatureReport command for issuing command.
2. Use GetFeatureReport command for getting results.
3. Use EP5 for data. Report size 64Byte.
4. Use EP6 for data. Report 64 Byte.
5. SetFeatureReport checksum 機制 没有使用校验和

   CS = (Byte0+Byte1+ ……+Byte6+Byte7) XOR (10101010B)

6.Vendor Page 0FF01H,Vendor Usage 022H.

✧Get firmware version (80H)： 讀去 F/W 版本

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	80	00	00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	80	H-Ver	L-Ver	mcu_type	sensor_type	key_sum	00	00

mcu_type:

mcu		备注
50343	01H	9
52341	02H	9
571	03H	7
572	04H	9
574	05H	9
553	06H	7
560	07H	9
541	08H	9
其他	FFH	

sensor_type:

sensor		备注
3050	01H	
3205	02H	
3325	03H	
3327	04H	
3336	05H	
3337	06H	
3360	07H	
3389	08H	
4090	09H	
OM36	21H	
OM37	22H	

✧Set current profile#(01H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	01	Profile#	00	00	00	00	00	CS

Profile range:0~5

✧Read current profile#(81H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	81	00	00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	81	Profile#	00	00	00	00	00	00

Profile#：当前的 profile

✧Set parameter_0(02H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	02	Profile#	00	00	00	00	01	CS
OUT(EP6)	total 64-byte parameter 0 data							

Profile# ： 当前的 profile，范围 0~5(返回大于 5 代表出错）

01：表示 data 需要的传输次数（每次 64byte）。

Data:

Byte0 ： reportrate-回报率(1ms 2ms 4ms 8ms)

Byte1 ： dpi_stage-DPI 段(0~7)

Byte2 ： ledEffect-灯效

Byte3 ： ledBrightness-亮度

Byte4 ： ledSpeed-速度

Byte5 ： ledColor-颜色

Byte6 ： ledOption-灯光流动方向

Byte7-byte63:保留

	Type	Brightness	Speed	Color	Option	备注
全灭	00H	xx	xx	xx	xx	
流光	01H	00--FF	01--07	0--7(7:loop)	0&1	
常亮	02H	00--FF	01--07	0--7(7:loop)	xx	
霓虹	03H	00--FF	01--07	0--7(7:loop)	xx	
呼吸	04H	00--FF	01--07	0--7(7:loop)	xx	
指动	05H	00--FF	01--07	0--7(7:loop)	xx	
雨滴	06H	00--FF	01--07	0--7(7:loop)	xx	
跑马	07H	00--FF	01--07	0--7(7:loop)	0&1	
点动	08H	00--FF	01--07	0--7(7:loop)	xx	

对碰	09H	00--FF	01--07	0--7(7:loop)	xx	
保留	0a~9FH					

✧Read parameter_0(82H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	82		00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	82	Profile#	00	00	00	00	01	00
IN(EP5)		total 64-byte parameter 0 data						

Profile# ： 当前的 profile，范围 0~5(返回大于 5 代表出错）

01：表示 data 需要的传输次数（每次 64byte）。

✧Set parameter_1(03H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	03	Profile#	index	00	00	status	times	CS
OUT(EP6)	total (64*3)byte parameter 1 data							

Profile# ： 当前的 profile，范围 0~5(返回大于 5 代表出错）

times：表示 data 需要的传输次数（每次 64byte）。最大 3*64=192 byte

index: 开始存储的数据索引(index*64)

Status: bit0-1 代表第 1 个 64BYTE 数据有改动

bit0-2 代表第 2 个 64BYTE 数据有改动

bit0-3 代表第 3 个 64BYTE 数据有改动

Data:

Byte0 ： btn_debounce- 按键消抖时间(单位：ms)最大不要超过 50ms

Byte1 ： wheel_debounce- 滚轮消抖时间（单位：ms）目前不使用

Byte2 ： sonser_config- sensor 相关设置 目前不使用

Byte3： sonser_lift- sensor 静默高度（2mm && 3mm）

Byte4 ： dpiStage_Max- DPI 段数（最大为 8） 目前不使用

Byte5~6： scale_x- DPI x 方向缩放比例（scale_x/100）目前不使用

Byte7~8： scale_y- DPI y 方向缩放比例（scale_y/100）目前不使用

Byte9~24 ： dpi_x_value[8] 8 段 DPI x 方向值（16 bit 长度，驱动不用做处理）

Byte0	Byte 1	……	Byte14	Byte15
value[0] 低位	value[0] 高位	……	value[7] 低位	value[7] 高位

Byte25~40 ： dpi_y_value[8] 8 段 DPI y 方向值

PS：同 X 方向

Byte41~63： reserve_var[23]- 保留 23 字节

//----------

Byte64~87: dpi RGB Color DPI 段数指示 RGB 颜色数据

Byte0	Byte 1	Byte2	……	Byte21	Byte22	Byte23
color[0][0]	color[0][1]	color[0][2]	……	color[7][0]	color[7][1]	color[7][2]

Byte88~111： LED_RGB_Color 灯效 RGB 颜色数据

PS:同 DPI RGB Color

Byte112~127： reserve_var[16]- 保留 16 字节

//----------

Byte128~191: KEY_maxtrix_table 按键定义数据（每个 KEY 用 4 个 byte 表示，共 16 个 KEY）

默认	Type
KEY0	鼠标左键
KEY1	鼠标右键
KEY2	鼠标中键
KEY3	鼠标 B4
KEY4	鼠标 B5
KEY5	保留
KEY6	保留
KEY7	保留
KEY7~15	保留

✧Read parameter_1(82H) //不使用

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	83		00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	83	Profile#	00	00	00	00	04	00
IN(EP5)		total (64*4)byte parameter 1 data						

Profile# ： 当前的 profile，范围 0~5(返回大于 5 代表出错）

04：表示 data 需要的传输次数（每次 64byte）。

✧Set macro key (04H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	04	Profile#	Macro#	00	00	00	02	CS

OUT(EP6)	total 128-byte macro key data for each

Macro# = 0~47.

✧Read macro key(84H)

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	84	00	00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	84	Profile#	Macro#	00	00	00	02	00
IN(EP5)	total (64*2)byte macro key data for each							

04：表示 data 需要的传输次数（每次 64byte）。

✧Set USB String (05H) //不使用

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	05	00	00	00	00	00	08	CS
OUT(EP6)	total (64*8)							

08：表示 data 需要的传输次数（每次 64byte）。

PS: Manufacture String 固定下 100byte，length 表示有效长度，无效 unicode 补 0 填充。

Product String 和 SerialNum 同理。例如：“ABC” length=3*2+2。3 个字符所以 3*2，加上 length 和 type,所以是+2.

Manufacture String, 100 byte，最大 49 個字

Manufacture String format :

Byte1	Byte2	Byte3	Byte4	Byte…	Byte98	Byte999
Length	Type	Unicode Char		….	Unicode Char	

Length = string length*2+2

Type = fixed 0x03

Product String, 100 byte，最大 49 個字

Product String format :

Byte1	Byte2	Byte3	Byte4	Byte…	Byte98	Byte99
Length	Type	Unicode Char		….	Unicode Char	

Length = string length*2+2

Type = fixed 0x03

SerialNumber String, 100byte，最大 49 個字

SerialNumber String format:

Byte1	Byte2	Byte3	Byte4	Byte…	Byte98	Byte99
Length	Type	Unicode Char		….	Unicode Char	

Length = string length*2+2

Type = fixed 0x03

日期, 100byte, 最大 49 個字

SerialNumber String format:

Byte1	Byte2	Byte3	Byte4	Byte…	Byte98	Byte99
Length	Type	Unicode Char		….	Unicode Char	

Length = string length*2+2

Type = fixed 0x03

✧Read USB String(85H) //不使用

	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
Setup	21	09	00	03	02	00	08	00
OUT	85	00	00	00	00	00	00	CS
Setup	A1	01	00	03	02	00	08	00
IN	85	00	00	00	00	00	08	00
IN(EP5)	total (64*8)							

二.Profile structure summary:

Key matrix (index table) structure:

There are at most 16 keys in gaming keyboard or mouse. Every key has corresponding 4-byte space that stores the key mode and the mapping data to the Host. Profile has an index table that records the function definition for each key. The firmware needs to read out the function definition of each key, and to parse by corresponding key mode. Then send out the right data sequence to the Host.

The format of the key index table is shown as below:

Byte3（4th）	Byte2（3rd）	Byte1（2nd）	Byte0（1st）
Null (00H)	--	--	--
KeyBoard(01H)	Code#2	Code#1	Code#0
Mouse Button(02H)	--	--	Code
Multi-media key(03H)	--	Code#1	Code#0
ACPI Key(04H)	--	--	Code
Macro Key(05H)	Macro Type	--	Macro Index
保留（06-09）			
LED Control(0AH)	--	--	Code#0
DPI settings(0BH)	Code#2	Code#1	Code#0
Fire Key(0CH)	Fire type	FireTimes	Code#0
保留（0F-7F）			
Special Key(F0H)	type	Code#1	Code#0

Vendor key(F1H)	Code#2	Code#1	Code#0

There are several key modes for the 4th byte of index table, as follow:

1. Keyboard key :
   For Keyboard key, the Key mode =01H, the 1st ,The 2nd and the 3rd byte indicate the keyboard key USB output key code (Please refer to attached table for keyboard key).

2. Mouse Button:
   For Mouse Button, the mode = 02H, the 1st byte indicate the button key USB output key code (Please refer to attached table for button key)。

3. Multi-media key:
   For Multi-media key, the Key mode = 03H, the multi-key key code is stored in the 1st byte and the 2nd byte respectively (Please refer to attached table for Multi-key).

4. ACPI key:
   For ACPI key, the Key mode = 04H, the 1st byte indicate the ACPI key USB output keycode (Please refer to attached table for ACPI key)。

5. Macro key :
   For Macro key, the key mode =05H, the 3rd byte represent macro key type, the2nd byte represent the index of the macro key. There is a table of index for all macro keys. The firmware will look up the corresponding macro key codes from the macro key definition buffer.

6. LED Control:
   For LED settings key, the Key mode = 0AH,fucnctions define as follows table:

3rd Byte	1st byte	Function
00	03H	灯效循环切换
	保留	

7. DPI switch key :
   For DPI switch key, the key mode = 0BH, fucnctions define as follows table:

1st byte	Function
01H	DPI+
02H	DPI-
03H	DPI+ loop

8. Fire Key :

Byte3	Byte2	Byte1	Byte0
0x0C	Repeat number	Delay	XX

XX 必須在 KEY_A(0x04) 跟 Mouse Scroll down(0xF8)之間.

Delay Time = Delay (1~255)* Time base(10ms)

Send Press →Delay * Time base →Send Release → Delay * Time base

Repeat number = 00, repeat stop for release button key

Repeat number = xx, repeat stop for counter repeat number

9. Special switch key :

For Special key, the key mode = F0H, fucnctions define as follows table:

3rd	2nd	1st byte	Function
00	--	--	Fn
01	--	01H	Profile number +
01	--	02H	Profile number -
01	--	03H	Profile number Loop
01	value	04H	Profile number value
02	--	01H	Report rate +
02	--	02H	Report rate -
02	--	03H	Report rate loop
02	value	03H	Report rate value
03	DPIValue_H	DPIValue_L	snipe
04	Delay time	Modify key	TAB Function

TAB Function

Byte1	Byte2	Byte3	Byte4
0xF0	04H	Delay time	Modify key

Modify key code = ALT(0xE2) or Win(0xE3).

Delay time (1~255)= Modify key Release Delay time (Time base = 10ms).

Send Modify+Tab → release Tab → delay →release modify key.

10. Vendor Hot key:

For Vendor Hot key, the Key mode = F1H.

三. Macro Key Data Format :

Macro Number 共有 48 個，每個 macro 為 250 Bytes，前 2 個 byte 為 macro code repeat counter,每 2 個 Bytes 為一個 action,當 2 個 byte 都為 00 時表示 macro code 結束,或是達到最後的 action。

Macro code type for Key matrix

Function	說明
0x00	1. Normal macro code,到 repeat counter 結束. 2. 有任何 button 按下既結束現有的 macro code.
0x01	1. Always repeat macro code,不理會 repeat counter 的值. 2. 有任何 button 按下既結束現有的 macro code.
0x02	1. Always repeat 到 Button Release 結束

There are long M * (2 + P*2) bytes buffer for all macro keys definitions. Every macro key has maxi. P key actions (P*2 bytes) and two bytes for repeat time. The macro buffer locate after the index buffer in the Flash. Every Macro key define as follow:.

a. Ability to define delay in milliseconds between keys' press-and-release
b. Example: Key "A" is defined as "Shift+W" 150 millisecond delay "Release W" 100 millisecond delay "Release Shift"
c. The mapping data is shown below:

Macro:

Byte1	Byte2
Repeat Low byte	Repeat High byte

where : This provide a range of values from 1 to 65535 repeat number.

Byte3	Byte4	Byte5	Byte6	….	Byte2*n	Byte2*n+1
Attribute1	Code No.1	Attribute2	Code No.2	….	Attribute n	Code No.n

Macro Key Example :

以下為一 Macro key 其內容為按下後送 123 重覆兩次，於 Byte 15 後無資料須全部補 00。

Byte1	Byte2
00	02

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12
01	1E	81	1E	01	1F	81	1F	01	20

Byte13	Byte14	Byte15	Byte16	..	..	..	..	Byte2*n	Byte2*n+1
81	20	00	00	..	..	..	..	00	00

Where:

1. Key Code No: refer to **Key Code in FLASH Table**
2. The content of macro key cannot be multi-media key.
3. Attribute: includes make/break key flag and delay time.
   Bit 7: 0 indicates **make** key; 1 indicates **break** key.
   Bit 0~6: indicate delay time. This provides a range of values from 0.01 to 1.27 seconds, with a 10 millisecond resolution. So really delay time = Delay Time * 10 millisecond. If the delay is equal to 0, then the next action.

Special key code	Function
0F5H	Send wheel data, next active for wheel value
0F6H	Send wheel data, next active for wheel value
0F7H	Send AC Pan, next active for AC Pan value
0F8H	Send AC Pan, next active for AC Pan value
0F9H	Send 8 bit X data ,next active for x value
0FAH	Send 8 bit Y data ,next active for y value

四. Vendor Code Definition:

Vendor code report format total 11 byte.

To send 11 byte data according the following data format to inform host

when Parameter change by the endpoint 2 of the inteface 1.(REPORT ID-0x05)

Byte0	Byte1	Byte2	Byte3	Byte 4	Byte 5	Byte 6
ID-0x04	Profile Number	reportrate	dpi_stage	ledEffect	保留	保留
Byte 7	Byte 8	Byte 9	Byte 10			
保留	保留	Code	Type			

Type	Code	说明
FEH	02H	Set param_0 succeed
FEH	03H	Set param_1 succeed
FEH	04H	Set macro succeed
FEH	05H	Set strings succeed
0xFF	00	当前 profile 已恢复默认设置
0xF1	Vendor code	AP 识别

五. Key Code Table:

		For Macro Key		For Key Matrix			
Key defined	**Key**	**Key mode**	**Key code**	**Byte3**	**Byte2**	**Byte1**	**Byte0**
No Event	Normal key	00	00(macro end)	00	00	00	00
Overrun Error	Normal key	00	01(delay only)				
POST Fail	Normal key	00	02(delay only)				
ErrorUndefined	Normal key	00	03(delay only)				
a A	Normal key	00	04	00	00	00	04
b B	Normal key	00	05	00	00	00	05
c C	Normal key	00	06	00	00	00	06
d D	Normal key	00	07	00	00	00	07
e E	Normal key	00	08	00	00	00	08
f F	Normal key	00	09	00	00	00	09
g G	Normal key	00	0A	00	00	00	0A

h H	Normal key	00	0B	00	00	00	0B
i I	Normal key	00	0C	00	00	00	0C
j J	Normal key	00	0D	00	00	00	0D
k K	Normal key	00	0E	00	00	00	0E
l L	Normal key	00	0F	00	00	00	0F
m M	Normal key	00	10	00	00	00	10
n N	Normal key	00	11	00	00	00	11
o O	Normal key	00	12	00	00	00	12
p P	Normal key	00	13	00	00	00	13
q Q	Normal key	00	14	00	00	00	14
r R	Normal key	00	15	00	00	00	15
s S	Normal key	00	16	00	00	00	16
t T	Normal key	00	17	00	00	00	17
u U	Normal key	00	18	00	00	00	18
v V	Normal key	00	19	00	00	00	19
w W	Normal key	00	1A	00	00	00	1A
x X	Normal key	00	1B	00	00	00	1B
y Y	Normal key	00	1C	00	00	00	1C
z Z	Normal key	00	1D	00	00	00	1D
1 !	Normal key	00	1E	00	00	00	1E
2 @	Normal key	00	1F	00	00	00	1F
3 #	Normal key	00	20	00	00	00	20
4 $	Normal key	00	21	00	00	00	21
5 %	Normal key	00	22	00	00	00	22
6 ^	Normal key	00	23	00	00	00	23
7 &	Normal key	00	24	00	00	00	24
8 *	Normal key	00	25	00	00	00	25
9 (	Normal key	00	26	00	00	00	26
0)	Normal key	00	27	00	00	00	27
Return	Normal key	00	28	00	00	00	28
Escape	Normal key	00	29	00	00	00	29
Backspace	Normal key	00	2A	00	00	00	2A
Tab	Normal key	00	2B	00	00	00	2B
Space	Normal key	00	2C	00	00	00	2C
- _	Normal key	00	2D	00	00	00	2D
= +	Normal key	00	2E	00	00	00	2E
[{	Normal key	00	2F	00	00	00	2F
] }	Normal key	00	30	00	00	00	30
\ |	Normal key	00	31	00	00	00	31
Europe 1 (Note 2)	Normal key	00	32	00	00	00	32
; :	Normal key	00	33	00	00	00	33
' "	Normal key	00	34	00	00	00	34
` ~	Normal key	00	35	00	00	00	35
, <	Normal key	00	36	00	00	00	36
. >	Normal key	00	37	00	00	00	37
/ ?	Normal key	00	38	00	00	00	38
Caps Lock	Normal key	00	39	00	00	00	39
F1	Normal key	00	3A	00	00	00	3A
F2	Normal key	00	3B	00	00	00	3B
F3	Normal key	00	3C	00	00	00	3C
F4	Normal key	00	3D	00	00	00	3D
F5	Normal key	00	3E	00	00	00	3E
F6	Normal key	00	3F	00	00	00	3F
F7	Normal key	00	40	00	00	00	40
F8	Normal key	00	41	00	00	00	41
F9	Normal key	00	42	00	00	00	42
F10	Normal key	00	43	00	00	00	43
F11	Normal key	00	44	00	00	00	44

F12	Normal key	00	45	00	00	00	45
Print Screen	Normal key	00	46	00	00	00	46
Scroll Lock	Normal key	00	47	00	00	00	47
Break (Ctrl-Pause)	Normal key	00	48	00	00	00	48
Pause	Normal key	00	48	00	00	00	48
Insert	Normal key	00	49	01	00	00	49
Home	Normal key	00	4A	00	00	00	4A
Page Up	Normal key	00	4B	00	00	00	4B
Delete	Normal key	00	4C	00	00	00	4C
End	Normal key	00	4D	00	00	00	4D
Page Down	Normal key	00	4E	00	01	01	4E
Right Arrow	Normal key	00	4F	00	00	00	4F
Left Arrow	Normal key	00	50	00	00	00	50
Down Arrow	Normal key	00	51	00	00	00	51
Up Arrow	Normal key	00	52	00	00	00	52
Num Lock	Normal key	00	53	00	00	00	53
Keypad /	Normal key	00	54	00	00	00	54
Keypad *	Normal key	00	55	00	00	00	55
Keypad -	Normal key	00	56	00	00	00	56
Keypad +	Normal key	00	57	00	00	00	57
Keypad Enter	Normal key	00	58	00	00	00	58
Keypad 1 End	Normal key	00	59	00	00	00	59
Keypad 2 Down	Normal key	00	5A	00	00	00	5A
Keypad 3 PageDn	Normal key	00	5B	00	00	00	5B
Keypad 4 Left	Normal key	00	5C	00	00	00	5C
Keypad 5	Normal key	00	5D	00	00	00	5D
Keypad 6 Right	Normal key	00	5E	00	00	00	5E
Keypad 7 Home	Normal key	00	5F	00	00	00	5F
Keypad 8 Up	Normal key	00	60	00	00	00	60
Keypad 9 PageUp	Normal key	00	61	00	00	00	61
Keypad 0 Insert	Normal key	00	62	00	00	00	62
Keypad . Delete	Normal key	00	63	00	00	00	63
Europe 2	Normal key	00	64	00	00	00	64
App	Normal key	00	65	00	00	00	65
Keyboard Power	Normal key	00	66	00	00	00	66
Keypad =	Normal key	00	67	00	00	00	67
F13	Normal key	00	68	00	00	00	68
F14	Normal key	00	69	00	00	00	69
F15	Normal key	00	6A	00	00	00	6A
F16	Normal key	00	6B	00	00	00	6B
F17	Normal key	00	6C	00	00	00	6C
F18	Normal key	00	6D	00	00	00	6D
F19	Normal key	00	6E	00	00	00	6E
F20	Normal key	00	6F	00	00	00	6F
F21	Normal key	00	70	00	00	00	70
F22	Normal key	00	71	00	00	00	71
F23	Normal key	00	72	00	00	00	72
F24	Normal key	00	73	00	00	00	73
Keypad , (Brazilian Keypad .)	Normal key	00	85	00	00	00	85
Keyboard Int'l 1 ろ (Ro)	Normal key	00	87	00	00	00	87
Keyboard Intl'2 かたかな ひらがな ローマ字 (Katakana/Hiragana)	Normal key	00	88	00	00	00	88

Keyboard Int'l 2 ¥ (Yen)	Normal key	00	89	00	00	00	89
Keyboard Int'l 4 前候補 変換 (次候補) 全候補 (Henkan)	Normal key	00	8A	00	00	00	8A
Keyboard Int'l 5 無変換 (Muhenkan)	Normal key	00	8B	00	00	00	8B
Keyboard Lang 1 한/영 (Hanguel/English)	Normal key	00	90	00	00	00	90
Left Control	Normal key	00	E0	00	00	00	E0
Left Shift	Normal key	00	E1	00	00	00	E1
Left Alt	Normal key	00	E2	00	00	00	E2
Left GUI	Normal key	00	E3	00	00	00	E3
Right Control	Normal key	00	E4	00	00	00	E4
Right Shift	Normal key	00	E5	00	00	00	E5
Right Alt	Normal key	00	E6	00	00	00	E6
Right GUI	Normal key	00	E7	00	00	00	E7
Mouse L Button	Mouse button	00	F0	02	00	00	F0
Mouse R Button	Mouse button	00	F1	02	00	00	F1
Mouse M Button	Mouse button	00	F2	02	00	00	F2
Mouse B4 Button	Mouse button	00	F3	02	00	00	F3
Mouse B5 Button	Mouse button	00	F4	02	00	00	F4
Mouse wheel up	Mouse button	00	F5	02	00	Value	F5
Mouse wheel dowm	Mouse button	00	F6	02	00	Value	F6
Mouse AC Pan left	Mouse button	00	F7	02	00	Value	F7
Mouse AC Pan right	Mouse button	00	F8	02	00	Value	F8
System Power	ACPI			04	00	00	01
System Sleep	ACPI			04	00	00	02
System Wake	ACPI			04	00	00	04
Scan Next Track	Multi-media key			03	00	00	B5
Scan Previous Track	Multi-media key			03	00	00	B6
Stop	Multi-media key			03	00	00	B7
Play/ Pause	Multi-media key			03	00	00	CD
Mute	Multi-media key			03	00	00	E2
Volume Up	Multi-media key			03	00	00	E9
Volume Down	Multi-media key			03	00	00	EA
Media Select	Multi-media key			03	00	02	83
Mail	Multi-media key			03	00	02	8A
Calculator	Multi-media key			03	00	02	92
My Computer	Multi-media key			03	00	02	94
WWW Search	Multi-media key			03	00	02	21
WWW Home	Multi-media key			03	00	02	23
WWW Back	Multi-media key			03	00	02	24
WWW Forward	Multi-media key			03	00	02	25
WWW Stop	Multi-media key			03	00	02	26
WWW Refresh	Multi-media key			03	00	02	27
WWW Favorites	Multi-media key			03	00	02	2A

Vendor code				F1	XX	YY	ZZ
Macro Key	Macro			05	Type	00	Index
DPI switch	DPI+			0B	00	00	01
DPI switch	DPI-			0B	00	00	02
DPI switch	DPI+ loop			0B	00	00	03
DPI switch	DPI Value			0B	Value_H	Value_L	04
Special Key	Fn			F0	00	00	00
Special Key	Profile+			F0	01	00	01
Special Key	Profile-			F0	01	00	02
Special Key	Profile+ loop			F0	01	00	03
Special Key	Profile value			F0	01	value	04
Special Keyh	Report rate +			F0	02	01	00
Special Key	Report rate -			F0	02	02	00
Special Key	Report rate + loop			F0	02	03	00
Special Key	Report rate Value			F0	02	Value	04
Special Key	snipe			F0	03	Value_H	Value_L
Special Key	Tab Function			F0	04	Delay time	Alt or Win
Fire Key	Fire mode			0C	repeat	delaytime	code
RGB Key	Effect Loop			0A	00	00	03
RGB Key	Effect Inc			0A	00	00	01
RGB Key	Effect Dec			0A	00	00	02
RGB Key	Effect Value			0A	00	Value	04
RGB Key	Brightness Loop			0A	01	00	03
RGB Key	Brightness Inc			0A	01	00	01
RGB Key	Brightness Dec			0A	01	00	02
RGB Key	Brightness Value			0A	01	Value	04
RGB Key	Speed Loop			0A	02	00	03
RGB Key	Speed Inc			0A	02	00	01
RGB Key	Speed Dec			0A	02	00	02
RGB Key	Speed Value			0A	02	Value	04
RGB Key	Color Loop			0A	03	00	03
RGB Key	Color Inc			0A	03	00	01
RGB Key	Color Dec			0A	03	00	02
RGB Key	Color Value			0A	03	Value	04
RGB Key	Option Loop			0A	04	00	03
RGB Key	Option Inc			0A	04	00	01
RGB Key	Option Dec			0A	04	00	02
RGB Key	Option Value			0A	04	Value	04